

RESEARCH ARTICLE

Evaluating COVID-19 risk under the estimation of population mean using two attributes

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Abstract

The virus of COVID-19 has affected humans physically, mentally, and economically all over the world. Each country's development level, resources, and immunization to have coping capacity against this covid19 differ country-wise. Thus, understanding socio-economic vulnerability and coping capacity among countries under different health systems can be crucial. Contrasting most articles on COVID-19, this article focuses on evaluating and estimating the COVID-19 risk and lack of coping capacity in 190 countries. This present study suggests an exponential estimator using two auxiliary attributes. Theoretically, the mean square error expressions are derived and compared with some existing estimators. These findings were supported by using the real data of INFORM COVID-19 risk.

KEYWORDS

auxiliary attributes, COVID-19 risk, exponential estimator, immunization coverage, lack of coping capacity, mean square error

1 | INTRODUCTION

The coronavirus, or COVID-19, is agitating panic for thus several reasons. Therefore, nobody has the immunity to fight it. Second, it is highly contagious, which means it spreads fast. Third, the scientists do not seem to be entirely sure of how it behaves since they need borderline history. Hoseinpour Dehkordi et al.¹ presented a new procedure based on a regression approach to estimate the COVID-19 case fatality rate during the pandemic. They have also compared the case history of a few Asian and European countries. Hoang² presented the characteristics of COVID-19 case detection in two phases to overcome the pandemic using the data set from the Ministry of Health of Vietnam. At that time, the situation in Vietnam was more volatile than expected. Overton et al.³ provided a toolkit to analyze the outbreak and access the interventions using the statistical and mathematical model-based toolkit. They applied three different COVID-19 pandemic datasets to test the toolkit and focused on parameter estimation under known biases in the datasets. Wong et al.⁴ explained that not only do different countries have different resources for the health system to make. The immunization is strong so that people can cope with the disease, but also have a diversity of economic, political, and social features that the government's effort was unable to cope with disasters like COVID-19. In an article by Silveira et al.⁵ the measures of relevance to psychological vulnerability, resilience, and social cohesion were identified for Germany in a heterogeneous Berlin population ranging from 18 to 65 years of age. Lockett et al.⁶ studied that people with higher levels of posttraumatic stress symptoms and COVID-19-related functional impairment would display greater anxiety and lower coping self-efficacy after pandemic reminders. From these studies, one can conclude that the COVID-19 pandemic devastated the people economically and physically and mentally in many cases.

In survey sampling, we deal with different interdisciplinary research territories such as population studies, agriculture, health issues, engineering, etc. Therefore, the researchers are interested in estimating the average using the proportion of the population that possesses a specified attribute as an auxiliary variable.

Wynn⁷ firstly suggested a ratio estimator using the auxiliary attribute when the population is divided into two classes. Singh et al.⁸ proposed a ratio of proportion utilizing the attribute as a piece of auxiliary information. Naik and Gupta⁹ discussed the point biserial correlation coefficient

idea. Solanki and Singh¹⁰ recommended a class of estimators for the population mean of the study variable using the known population proportion of the auxiliary attribute. Malik and Singh¹¹ proposed the exponential type of estimator using two auxiliary attributes. Zaman¹² suggested a new family of estimators using auxiliary information for simple random sampling when the data are qualitative in nature. Yadav and Zaman¹³ presented a family of estimators of population mean utilizing the known parameters of the auxiliary attribute. Zaman,¹⁴ Sajjad et al.,¹⁵ Zaman and Kadilar,¹⁶ and Bhushan et al.¹⁷ suggested estimators estimate the population mean using auxiliary attributes. Bhushan et al.¹⁸ introduced a modified class of estimators to estimate the population variance using auxiliary attributes. The auxiliary attributes are the information readily available sometimes to estimate the variable of interest for example: to estimate the proportion of voters favoring a proposition, the proportion of drug usage in both genders, the irrigation status, and yield of the crop, etc.

Regarding the COVID-19 pandemic, few survey statisticians have participated in evaluating and estimating the dynamics of the evolution of COVID-19 risk as Shahzad et al.¹⁹ presented some new calibration estimators to estimate population variance of recovery time of COVID-19 under two-phase stratified sampling design. Further, Zaman et al.¹⁴ presented two families of multivariate exponential estimators for simple random sampling to measure the proportion of health care and socioeconomic vulnerability during the COVID-19 pandemic.

This present study estimated the lack of coping capacity among different countries of the world based on their health systems and immunization capability using the INFORM COVID-19 risk datasets. The INFORM COVID-19 risk index facilitates us with the devastating effects of the pandemic on the human health system all over the world. This article aims to suggest a general form of the exponential estimator to estimate the population mean for two auxiliary attributes using COVID-19 risk. The expression of the mean square error (MSE) of the suggested estimator is derived up to the first order of approximation. The outline of the article is organized as follows: In Section 2 the suggested estimator is introduced using the two auxiliary attributes along with the MSE expression. In Section 3, the theoretical comparison of the suggested exponential estimator and some existing estimators using the MSE expressions are reported. A real-life application using INFORM COVID-19 risk dataset is discussed in Section 4. The concluding remarks are given in Section 5.

2 | METHODOLOGY

In this section properties of the proposed estimator are explained, and datasets are described.

2.1 | The suggested estimator and properties

The exponential estimators can be used for the exponential relationship between study and auxiliary attributes Sanaullah et al.²⁰ The current article aims to propose an exponential estimator to estimate the population mean of the study variable considering the information about the population proportion possessing certain attributes.

2.1.1 | Notations

Consider $U = (u_1, u_2, \dots, u_N)$ be a finite population of size N . We draw a sample of size n by SRSWOR from a population U . Let y_i be the study variable and ϕ_i be the characteristic of the auxiliary attribute such as:

$\phi_i = 1$ if the i th unit possesses the attribute.

$\phi_i = 0$ other.

Let $A = \sum_{i=1}^N \phi_i$ and $a = \sum_{i=1}^n \phi_i$ be total units in population and sample, respectively.

Let $P = A/N$ and $p = a/n$ be the proportion of units in the population and sample, respectively.

Let $\bar{Y} = \frac{\sum_{i=1}^N y_i}{N}$ and $\bar{y} = \frac{\sum_{i=1}^n y_i}{n}$ be the population and sample mean, respectively.

Let P_1 and P_2 be the population proportion and p_1 be the sample proportion of the auxiliary attributes.

To obtain MSE the error term and their expectations are given as:

$$e_o = \frac{(\bar{y} - \bar{Y})}{\bar{Y}}, e_1 = \frac{(p_1 - P_1)}{P_1}, e_2 = \frac{(p_2 - P_2)}{P_2} \text{ be the error terms.}$$

such that.

$$E(e_i) = 0, (i = 0, 1, 2), E(e_0^2) = \lambda C_Y^2, E(e_1^2) = \lambda C_{p1}^2, E(e_2^2) = \lambda C_{p2}^2,$$

$$E(e_0 e_1) = \lambda \rho_{yp1} C_Y C_{p1}, E(e_0 e_2) = \lambda \rho_{yp2} C_Y C_{p2}, E(e_1 e_2) = \lambda \rho_{p1p2} C_{p1} C_{p2},$$

$$\text{where } \lambda = \frac{1}{n} - \frac{1}{N}, \rho_{yp1} = \frac{S_{yp1}}{S_Y S_{p1}}, \rho_{yp2} = \frac{S_{yp2}}{S_Y S_{p2}}, \rho_{p1p2} = \frac{S_{p1p2}}{S_{p1} S_{p2}},$$

$$\begin{aligned} C_Y &= \frac{S_Y}{Y}, C_{p1} = \frac{S_{p1}}{P_1}, C_{p2} = \frac{S_{p2}}{P_2}, S_Y^2 = \frac{\sum_{i=1}^N (y_i - \bar{Y})^2}{N-1}, S_{p1}^2 = \frac{\sum_{i=1}^N (p_{1i} - P_1)^2}{N-1} \\ S_{p2}^2 &= \frac{\sum_{i=1}^N (p_{2i} - P_2)^2}{N-1}, S_{yp1} = \frac{\sum_{i=1}^N (y_i - \bar{Y})(p_{1i} - P_1)}{N-1}, S_{yp2} = \frac{\sum_{i=1}^N (y_i - \bar{Y})(p_{2i} - P_2)}{N-1} \\ S_{p1p2} &= \frac{\sum_{i=1}^N (p_{1i} - P_1)(p_{2i} - P_2)}{N-1}. \end{aligned}$$

2.1.2 | Properties of the suggested estimator

We propose the following unbiased exponentiation ratio type estimator in estimating the population mean when using two auxiliary attributes as given by.

$$\bar{y}_M = \sum_{m=0}^3 \alpha_m t_m \quad (m = 0, 1, 2, 3). \quad (1)$$

Such that $\alpha_m \in R$, and α_m ($m = 0, 1, 2, 3$) denotes suitably chosen constants and R stands for the set of real numbers.

Rewriting (1), we have

$$\bar{y}_M = \alpha_0 t_0 + \alpha_1 t_1 + \alpha_2 t_2 + \alpha_3 t_3, \quad (2)$$

$$\text{where } t_0 = \bar{y}, t_1 = \bar{y} \exp\left(\frac{P_1 - p_1}{P_1 + p_1}\right), t_2 = \bar{y} \exp\left(\frac{P_2 - p_2}{P_2 + p_2}\right), t_3 = \bar{y} \exp\left(\frac{P_1 - p_1}{P_1 + p_1}\right) \exp\left(\frac{P_2 - p_2}{P_2 + p_2}\right).$$

then $\bar{y}_M$ can be written as:

$$\bar{y}_M = \bar{y} \left[\alpha_0 + \alpha_1 \exp\left(\frac{P_1 - p_1}{P_1 + p_1}\right) + \alpha_2 \exp\left(\frac{P_2 - p_2}{P_2 + p_2}\right) + \alpha_3 \exp\left(\frac{P_1 - p_1}{P_1 + p_1}\right) \exp\left(\frac{P_2 - p_2}{P_2 + p_2}\right) \right]. \quad (3)$$

Expressing the above equation in terms of error terms to a first-order approximation, we have

$$\bar{y}_M = \bar{y} (1 + e_0) \left[\alpha_0 + \alpha_1 \exp\left(\frac{-e_1}{2 + e_1}\right) + \alpha_2 \exp\left(\frac{-e_2}{2 + e_2}\right) + \alpha_3 \exp\left(\frac{-e_1}{2 + e_1}\right) \exp\left(\frac{-e_2}{2 + e_2}\right) \right].$$

The MSE of the above equation is given as:

$$\text{MSE}(\bar{y}_M) = \bar{Y}^2 E \left[e_0 - e_1 \left(\frac{\alpha_1}{2} + \frac{\alpha_3}{2} \right) - e_2 \left(\frac{\alpha_2}{2} + \frac{\alpha_3}{2} \right) + O(e^2) \right]^2. \quad (5)$$

Let

$$\begin{aligned} \frac{\alpha_1}{2} + \frac{\alpha_3}{2} &= \alpha' \text{ (say)} \\ \frac{\alpha_2}{2} + \frac{\alpha_3}{2} &= \alpha^* \text{ (say)}, \end{aligned} \quad (6)$$

where α' and α^* are another constant.

Therefore MSE of $\bar{y}_M$ in the form of α' and α^* can be written as

$$\text{MSE}(\bar{y}_M) \cong \bar{Y}^2 E(e_o - \alpha' e_1 - \alpha^* e_2)^2. \quad (7)$$

By expanding and applying expectations, we get

$$\text{MSE}(\bar{y}_M) = \theta \bar{Y}^2 \left[C_y^2 + \alpha'^2 C_{p1}^2 + \alpha^{*2} C_{p2}^2 - 2\alpha' \rho_{yp1} C_y C_{p1} - 2\alpha^* \rho_{yp2} C_y C_{p2} + 2\alpha' \alpha^* \rho_{p1p2} C_{p1} C_{p2} \right]. \quad (8)$$

Now setting

$$\frac{\partial \text{MSE}(\bar{y}_M)}{\partial \alpha'} = 0$$

So $\rho_{yp1} = \rho_{yp2}$.

Let $L_{ab} = \rho_{ab} \frac{C_a}{C_b}$.

The subscripts a and b show the presence of classes and $a \neq b$.

Therefore $\alpha' = L_{yp1} - \alpha^* L_{p1p2}$.

Now $\frac{\partial \text{MSE}(\bar{y}_M)}{\partial \alpha^*} = 0$, we get

$$\alpha^* = L_{yp2} - \alpha' L_{p1p2},$$

where $L_{yp2} = \rho_{yp2} \frac{C_y}{C_{p2}}$ and $L_{p1p2} = \rho_{p1p2} \frac{C_{p1}}{C_{p2}}$.

Now solving the above, we have

$$\left. \begin{aligned} \alpha' &= \frac{L_{yp1} - L_{yp2} L_{p1p2}}{1 - L_{p1p2} L_{p2p1}} = L(\text{say}) \\ \alpha^* &= \frac{L_{yp2} - L_{yp1} L_{p1p2}}{1 - L_{p1p2} L_{p2p1}} = L^*(\text{say}) \end{aligned} \right\}. \quad (9)$$

Substituting α' and α^* after solving we get the minimum given by

$$\begin{aligned} \text{MSE}(\bar{y}_M)_{\min} &\cong \theta \bar{Y}^2 C_y^2 \left(1 - \frac{\rho_{yp1}^2 + \rho_{yp2}^2 - 2\rho_{yp1}\rho_{yp2}\rho_{p1p2}}{(1 - \rho_{p1p2}^2)} \right) \\ \text{MSE}(\bar{y}_M)_{\min} &\cong \theta \bar{Y}^2 C_y^2 (1 - R_{y,p1p2}^2), \end{aligned} \quad (10)$$

where

$$R_{y,p1p2}^2 = \frac{\rho_{yp1}^2 + \rho_{yp2}^2 - 2\rho_{yp1}\rho_{yp2}\rho_{p1p2}}{(1 - \rho_{p1p2}^2)}.$$

3 | COMPARISON OF THE ESTIMATORS

This section compares the efficiencies of the proposed estimator with other estimators based on their MSE expressions. In the first part, we discussed existing estimators and in the second part, we make the theoretical comparison.

3.1 | Existing estimators

i. The usual mean estimator is

$$\bar{y}_0 = \bar{y}. \quad (11)$$

The var of the $\bar{y}_0$ is given as:

$$\text{Var}(\bar{y}_0) = \theta \left[\bar{Y}^2 C_y^2 \right]. \quad (12)$$

ii. The regression type estimator is given as:

$$\bar{y}_{\text{Reg}} = \bar{y} + \beta(P - p) \quad (13)$$

where β is given as

$$\beta = \frac{\text{Cov}(\bar{Y}, P)}{\text{Var}(P)},$$

$$\text{Var}(\bar{y}_{\text{Reg}}) = \theta \bar{Y}^2 C_y^2 [1 - \rho_{yp}^2], \quad (14)$$

iii. The exponential type of estimator is given as:

$$\bar{y}_{\text{exp}} = \bar{y} \exp\left(\frac{P - p}{P + p}\right), \quad (15)$$

$$\text{MSE}(\bar{y}_{\text{exp}}) = \theta \bar{Y}^2 \left[C_y^2 + \frac{1}{4} C_p^2 - 2\rho_{yp} C_y C_p \right]. \quad (16)$$

iv. Kumar and Bhoughal proposed exponential ratio-product estimator is given as:

$$\bar{y}_{\text{KB}} = \bar{y} \left[\alpha \exp\left(\frac{P_1 - p_1}{P_1 + p_1}\right) + (1 - \alpha) \exp\left(\frac{p_1 - P_1}{p_1 + P_1}\right) \right]. \quad (17)$$

$$\text{MSE}(\bar{y}_{\text{RP}})_{\min} = \theta \bar{Y}^2 C_y^2 [1 - \rho_{yp_1}^2]. \quad (18)$$

The optimum value of α is given as:

$$\alpha_{\text{opt}} \approx \frac{1}{2} + \rho_{yp_1} \frac{C_y}{C_{p1}}.$$

v. Singh and Kumar suggested a ratio-product estimator for two auxiliary attributes given:

$$\bar{y}_{\text{SK}} = \bar{y} \frac{P_1}{p_1} \frac{P_2}{p_2}. \quad (19)$$

$$\text{MSE}(\bar{y}_{\text{SK}}) = \theta \bar{Y}^2 \left[C_y^2 + C_{p_1}^2 + C_{p_2}^2 - 2(\rho_{yp_1} C_y C_{p_1} + \rho_{yp_2} C_y C_{p_2} - \rho_{p_1 p_2} C_{p_1} C_{p_2}) \right]. \quad (20)$$

3.2 | Theoretical comparison of the proposed and existing estimators

i. From (12) and (10), we have

$$\text{Var}(\bar{y}_M)_{\min} \leq \text{Var}(\bar{y}_0) = R_{y,p1p2}^2 \geq 0, \quad (21)$$

$$\text{where } R_{y,p1p2}^2 = \frac{\rho_{yp1}^2 + \rho_{yp2}^2 - 2\rho_{yp1}\rho_{yp2}\rho_{p1p2}}{(1 - \rho_{p1p2}^2)}.$$

ii. From (14) and (10), we have

$$\text{Var}(\bar{y}_M)_{\min} \leq \text{Var}(\bar{y}_{\text{Reg}}) = R_{y,p1p2}^2 - \rho_{yp1}^2 \geq 0. \quad (22)$$

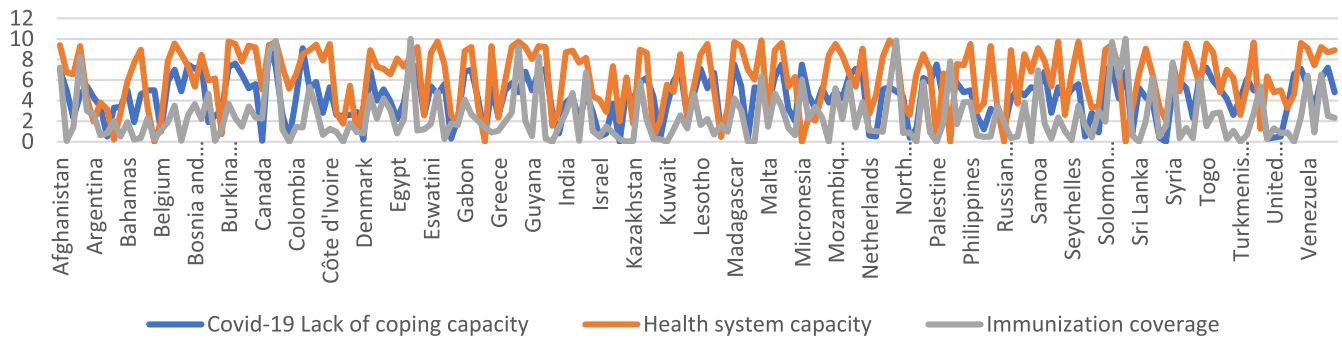


FIGURE 1 COVID-19 lack coping capacity with health system capacity and immunization coverage for 190 countries

iii. From (16) and (10), we have

$$\text{Var}(\bar{y}_M)_{\min} \leq \text{Var}(\bar{y}_{\text{exp}}) = \left[\frac{1}{4} C_p^2 + 2\rho_{yp} C_y C_p + C_y^2 R_{y,p1p2}^2 \right] \geq 0. \quad (23)$$

iv. From (18) and (10), we have

$$\text{var}(\bar{Y}_M)_{\min} \leq \text{var}(\bar{Y}_{KB}) = R_{y,p1p2}^2 - \rho_{yp1}^2 \geq 0. \quad (24)$$

v. From (20) and (10), we have

$$\begin{aligned} \text{var}(\bar{Y}_M)_{\min} &\leq \text{var}(\bar{Y}_{SK}). \\ [C_{p1}^2 + C_{p2}^2 - 2(\rho_{yp1} C_y C_{p1} + \rho_{yp2} C_y C_{p2} - \rho_{p1p2} C_{p1} C_{p2}) + C_y^2 R_{y,p1p2}^2] &\geq 0 \end{aligned} \quad (25)$$

The proposed estimators are more efficient than traditional estimators if conditions (21) to (25) are satisfied.

4 | THE REAL-LIFE EXAMPLE

In this section, the performance of the proposed estimator has been evaluated by using INFORM COVID-19 risk data.

4.1 | The data description

The data concerning the COVID-19 risk index, access to the socioeconomic vulnerability index, and lack of coping capacity were obtained from the INFORM COVID-19 risk dataset.²¹ In this article, different studies and auxiliary attributes have been chosen. Firstly, the study variable is the lack of coping capacity data scaled for 190 countries. From the same index, the first auxiliary attribute P1 is health system capacity data scaled from 0 to 10 (0 is *low capacity*, 10 is *high capacity*). The other auxiliary attribute P2 is immunization coverage based on a scale of 0 to 10 (0 is *low coverage*, 10 is *high coverage*).

Secondly, from the socioeconomic vulnerability index, the study variable is the COVID-19 vulnerability and attribute variables are P1 is awareness, and P2 is demographic and co-morbidities. The figures for datasets are given as:

According to Figure 1, Canada, Denmark, and Syria have the least COVID-19 lack coping capacity whereas Colombia is a highly lacking country with coping capacity for COVID-19. If we observe health system capacity, Belgium and Micronesia have the least proportion as compared to Malta with 10. Similarly, we noticed the immunization proportion coverage among 190 countries with which Egypt and Sri Lanka with 10.1 which means the highest immunization coverage capacity.

In Figure 2, comparing 190 countries Peru and Denmark have the highest vulnerability capacity of bearing COVID-19 risk. If we observe the same figure, Canadian people are more aware as compared to all other countries, and the United States has the least awareness regarding COVID-19. Burkina Faso has the uppermost demographic and co-morbidities proportion as compared to the other countries.

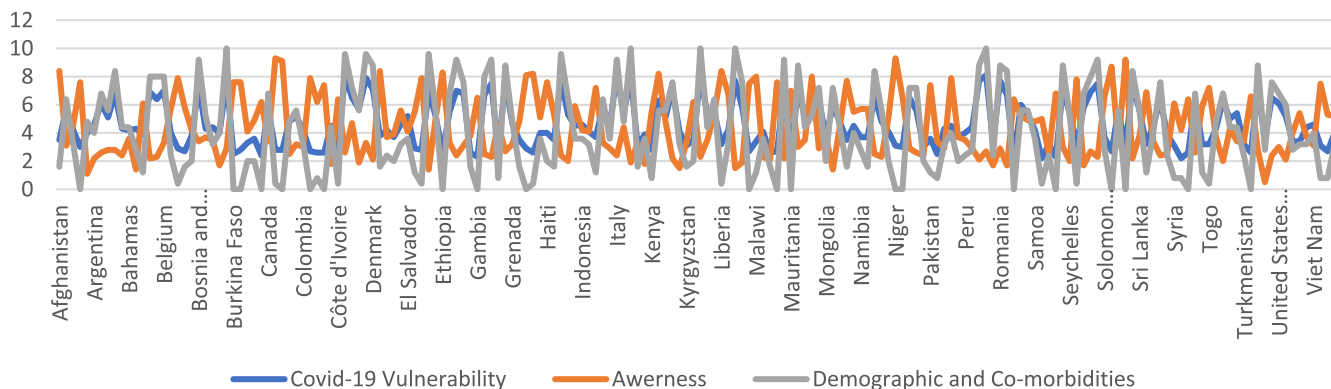


FIGURE 2 COVID-19 vulnerability evaluate with awareness and demographic and co-morbidities

TABLE 1 Mean square error (MSE) and PRE for dataset 1

Estimators	n = 60		n = 80		n = 100	
	MSE	PRE	MSE	PRE	MSE	PRE
$\bar{y}_0$	0.05455	100	0.03433	100	0.02276	100
$\bar{y}_{Reg}$	0.05441	100.25	0.03421	100.35	0.02266	100.44
$\bar{y}_{exp}$	0.01214	449.34	0.00764	449.36	0.00505	450.69
$\bar{y}_M$	0.00639	853.67	0.00402	853.98	0.00266	855.63
y_{KB}	0.00653	835.37	0.00413	831.23	0.00291	782.13
y_{SK}	0.00805	677.63	0.00632	543.19	0.00381	597.37

TABLE 2 Mean square error (MSE) and PRE for dataset 2

Estimators	n = 60		n = 80		n = 100	
	MSE	PRE	MSE	PRE	MSE	PRE
$\bar{y}_0$	0.00059	100	0.00043	100	0.00033	100
$\bar{y}_{Reg}$	0.00051	115.68	0.00032	134.37	0.00021	157.14
$\bar{y}_{exp}$	0.00030	196.67	0.00028	215.00	0.00016	206.25
$\bar{y}_M$	0.00027	218.51	0.00019	226.31	0.00009	366.67
y_{KB}	0.00028	210.71	0.00021	204.76	0.00010	330.00
y_{SK}	0.00029	203.44	0.00025	172.00	0.00013	253.84

4.2 | Results and discussion

The MSE and percent relative efficiency of the proposed and exponential estimator with some existing estimators for $N = 190$ countries are presented in Tables 1 and 2. Using simple random sampling, we assume $n = 60$, $n = 80$, and $n = 100$ for two datasets.

The relative efficiency values of the proposed estimator concerning traditional estimators are as below.

$$PRE = \frac{MSE(\bar{y}_0)}{MSE(\bar{y}_i)} \quad (26)$$

$i = \text{Reg, exp, M, KB, SK.}$

As noted earlier in Table 1, Y represents lack of coping capacity, P_1 means the proportion of health capacity, and P_2 represents the proportion of immunization coverage considered for the present study. The coefficient of correlation between lack of coping capacity and proportion of health

capacity is 0.1484. These values indicate that the lack of coping capacity during COVID-19 and the proportion of health capacity have a very low correlation. In other words, the lack of coping capacity is not attentively related to the proportion of health capacity. Similarly, for the lack of coping capacity and the proportion of immunization coverage correlation coefficient is 0.4731. This correlation coefficient value specifies that the two variables are moderately correlated; in other words, the lack of coping capacity is thoroughly correlated to immunization coverage.

In Table 2, the study variable Y symbolizes COVID-19 vulnerability, P_1 signifies the proportion of awareness about COVID-19, and P_2 represents the proportion of demographic and co-morbidities during COVID-19 considered for the present study. The correlation coefficient value between COVID-19 vulnerability and the proportion of awareness is -0.590 . It shows that the variables COVID-19 vulnerability and the proportion of awareness are strictly contradictory with a negative correlation. On the other hand, the coefficient of correlation between the variables COVID-19 vulnerability and the proportion of demographic and co-morbidities is 0.3326 . This value indicates that both variables are moderately related to each other.

In this context, a generalized exponential estimator has been proposed to estimate the population mean of study variable having two different situations of COVID-19 and their attributes such as (i) Y represents lack of coping capacity, P_1 represents the proportion of health capacity, and P_2 represents the proportion of immunization coverage (ii) Y symbolizes COVID-19 vulnerability, P_1 signifies the proportion of awareness about COVID-19, P_2 represents the proportion of demographic and co-morbidities during COVID-19. Using these two datasets, the proposed estimator and some existing estimators are compared numerically and theoretically. Tables 1 and 2 represent the MSE and PRE results of the proposed exponential estimator and some existing estimators. It is noticed from the numerical results that using both datasets the generalized exponential estimator $\bar{y}_M$ performs more efficiently as compared to the competing estimators either for positive or negative correlation among study and auxiliary attributes.

5 | CONCLUSION

This study proposes an exponential estimator to estimate the population mean using two auxiliary attributes performing more efficiently as compared to existing estimators mentioned in this article. The statistical properties of the estimator such as the bias and minimum MSE have been derived in the first order of approximation. In the first step, the proposed estimator is theoretically compared with the existing estimators. After that COVID-19 risk dataset is used in the numerical comparison. The numerical study confirms that the proposed estimator has the minimum MSE and maximum PRE values among compared estimators. Therefore, using the proposed estimators in practice for the present study and such issues is recommended.

CONFLICT OF INTEREST

The authors have no conflicts of interest to declare. All coauthors have seen and agree with the contents of the manuscript and there is no financial interest to report. We certify that the submission is original work and is not under review at any other publication.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available on <https://drmkc.jrc.ec.europa.eu/inform-index/INFORM-Covid>. The code that supports the findings of this study is available at <https://drmkc.jrc.ec.europa.eu/inform-index>.

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